



Physical Security Risk Assessment Methodology

LEO Basestation Ground Infrastructure

Classification: **CLIENT CONFIDENTIAL**

Version: 1.0 | March 2026

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1. Executive Summary

This document establishes a repeatable, transparent methodology for assessing the physical security risk to Low Earth Orbit (LEO) satellite basestation ground infrastructure. While the cyber threat landscape for LEO constellations has been extensively studied, the physical security of ground segment assets remains under-examined despite representing a critical single point of failure in constellation architectures.

The methodology is designed for application across all major LEO constellation operators, including SpaceX Starlink, Amazon Project Kuiper, Eutelsat OneWeb, and Telesat Lightspeed. It provides a structured framework that can be applied by Concentric field assessors to produce standardised, comparable risk ratings across a global portfolio of basestation sites.

The accompanying spreadsheet provides an initial risk register of identified global LEO basestation locations with preliminary risk ratings derived from this methodology.

2. Scope and Applicability

This methodology addresses physical security threats to LEO basestation ground infrastructure exclusively. It does not replicate the digital/cyber risk frameworks already established by NIST CSF alignment, NTRA regulatory compliance, or operator-specific cybersecurity programmes. The focus is on threats that manifest through physical access, proximity, or kinetic action against ground segment assets.

2.1 Assets in Scope

- Gateway Earth Stations (GES): Primary ground stations hosting Ka/Ku-band phased array antenna arrays, radome enclosures, and fibre interconnects to Internet Exchange Points.
- Satellite Network Portals (SNP): Data processing and routing facilities co-located with or adjacent to gateway stations.
- Telemetry, Tracking and Command (TT&C) Facilities: Sites responsible for satellite orbit management and health monitoring.
- Supporting Infrastructure: Power substations, backup generators, cooling systems, fibre landing points, and access roads serving basestation complexes.

2.2 Assets Out of Scope

- User terminal equipment and residential/commercial customer premises.
- Space segment assets (satellites, inter-satellite links).
- Mission Operations Centres (MOCs) unless co-located with ground stations.

3. Physical Threat Taxonomy for LEO Ground Stations

The following threat categories have been developed specifically for LEO ground infrastructure, informed by the segmented risk model established in the source reference material and adapted to address physical rather than digital vectors.

Threat Category	Threat ID	Description
Unauthorised Access	PA-01	Perimeter breach by hostile actors seeking to damage, destroy, or tamper with antenna arrays, radomes, or data centre equipment.
	PA-02	Insider threat from maintenance contractors, construction workers, or co-located facility staff with legitimate physical access.
Sabotage / Vandalism	PS-01	Deliberate damage to antenna infrastructure, fibre optic cables, or power supply systems by ideologically motivated, criminal, or state-sponsored actors.
	PS-02	Destruction of access roads or supporting utilities to deny operational capability.
Surveillance / Reconnaissance	PR-01	Hostile reconnaissance of site layout, security posture, shift patterns, and vulnerability mapping by adversary actors.
Theft	PT-01	Theft of high-value electronic components, copper cabling, or backup power equipment.
Drone / UAS Threat	PD-01	Unmanned aerial systems used for surveillance, signal interference, or direct kinetic attack on exposed antenna arrays.
Environmental / Natural Hazard	PE-01	Severe weather, seismic events, flooding, wildfire, or volcanic activity affecting site operability and structural integrity.
Civil Unrest / Protest	PC-01	Protest activity, blockade, or civil disorder targeting the facility or its operator due to political, environmental, or social grievance.
State-Level Threat	PX-01	Covert or overt state action against ground infrastructure in the context of geopolitical competition, hybrid warfare, or conflict escalation.

4. Assessment Methodology: The Concentric 5-Phase Process

The methodology follows a structured five-phase process aligned with the Orient-Assess-Control-Conform model referenced in NTRA regulatory frameworks, adapted for physical security application.

4.1 Phase 1: Orient

Establish the assessment scope by identifying and cataloguing all physical assets at the target site. This phase produces a Site Asset Register including GPS coordinates, building footprints, antenna counts, perimeter dimensions, access points, and utility interconnections. The assessor maps the site against the asset categories defined in Section 2.1.

4.2 Phase 2: Context

Evaluate the environmental and geopolitical context of the site. This phase produces a Site Context Profile covering seven scoring dimensions, each rated 1 to 5. These dimensions form the core of the risk calculation and are detailed in Section 5.

4.3 Phase 3: Threat Assessment

Apply the Threat Taxonomy (Section 3) to the specific site, assessing the likelihood and consequence of each applicable threat. Likelihood is informed by the Context Profile scores and local intelligence. Consequence is assessed against a standardised impact scale covering operational disruption, financial loss, reputational damage, and national security implications.

4.4 Phase 4: Vulnerability Assessment

Conduct a detailed assessment of existing physical security controls against each identified threat. This includes perimeter security (fencing, barriers, detection systems), access control (authentication, visitor management, lock hardware), surveillance systems (CCTV, monitoring coverage, retention), guard force (manning levels, response times, training standards), and resilience measures (backup power, redundant connectivity, hardening against natural hazards).

4.5 Phase 5: Risk Rating and Recommendations

Integrate findings from Phases 2 through 4 to produce a composite Physical Security Risk Rating for the site. Apply the risk matrix (Section 6) and generate prioritised recommendations with implementation timelines and cost estimates.

5. Scoring Dimensions: The Context Profile

Each site is assessed against seven dimensions. Each dimension is scored 1 (lowest risk) to 5 (highest risk). The composite Context Score is the weighted average of all dimensions.

Dimension	Weight	Assessment Criteria	Score Range
Geopolitical Stability	20%	Country-level stability, conflict proximity, sanctions exposure, governance quality (Fragile States Index, EIU Democracy Index).	1 = Stable NATO/EU state; 5 = Active conflict zone
Local Crime Environment	15%	Violent crime rates, organised crime presence, theft/burglary rates within 50km radius of site.	1 = Very low crime; 5 = High violent crime
Protest / Activism Risk	10%	History of anti-technology, environmental, or anti-corporate protest activity targeting communications infrastructure.	1 = No history; 5 = Active / recent targeting
Natural Hazard Exposure	15%	Seismic zone classification, flood plain mapping, hurricane/cyclone exposure, wildfire risk, extreme temperature events.	1 = Minimal hazards; 5 = Multiple severe hazards
Site Isolation / Remoteness	15%	Distance to nearest law enforcement, emergency services response time, road accessibility, proximity to population centres.	1 = Urban, rapid response; 5 = Extremely remote
Existing Security Posture	15%	Observed or reported quality of perimeter security, access control, CCTV, guard force, and hardening measures.	1 = Robust controls; 5 = Minimal/no controls
Strategic Value / Criticality	10%	Constellation dependency on this node (redundancy level), military/government use, TT&C function, population served.	1 = High redundancy; 5 = Single point of failure

Composite Context Score = $(\text{Geopolitical} \times 0.20) + (\text{Crime} \times 0.15) + (\text{Protest} \times 0.10) + (\text{Natural Hazard} \times 0.15) + (\text{Isolation} \times 0.15) + (\text{Security Posture} \times 0.15) + (\text{Strategic Value} \times 0.10)$

6. Risk Matrix

The final Physical Security Risk Rating for each site is determined by plotting Likelihood (derived from threat assessment and context scoring) against Consequence (derived from impact analysis).

Likelihood \ Consequence	Negligible	Low	Moderate	High
Very Likely	Moderate	High	Critical	Critical
Likely	Low	Moderate	High	Critical
Possible	Low	Low	Moderate	High
Unlikely	Negligible	Low	Low	Moderate
Rare	Negligible	Negligible	Low	Low

Risk Level	Response Requirement
Negligible	Monitor. No additional physical security measures required beyond baseline standards.
Low	Awareness. Standard perimeter and access control measures. Annual reassessment.
Moderate	Enhanced controls. Active CCTV monitoring, intrusion detection, regular patrol. Semi-annual reassessment.
High	Significant investment. 24/7 guard force, hardened perimeter, integrated alarm systems, liaison with local law enforcement. Quarterly reassessment.
Critical	Maximum protection. Armed security, counter-UAS capability, blast protection, redundant site planning, government coordination. Continuous monitoring.

7. Application to LEO Ground Segment Architecture

The physical risk profile of a LEO basestation is shaped by factors unique to the satellite communications ground segment. Assessors must account for the following architectural considerations when applying this methodology.

7.1 Transparent vs. Regenerative Architecture Implications

In transparent (bent-pipe) architectures, the ground station hosts the entire gNB processing stack, making it an exceptionally high-value target. Loss of a ground station in a transparent architecture directly eliminates connectivity for all users in the coverage footprint. In regenerative architectures with inter-satellite links (ISLs), the satellite can route traffic via alternative ground stations, reducing single-point-of-failure risk but not eliminating the physical threat to the node itself.

Assessors must determine the architecture type served by each ground station and adjust the Strategic Value / Criticality score accordingly. Transparent-architecture stations warrant a minimum score of 3 on this dimension.

7.2 Handover Topology and Redundancy

LEO constellations depend on continuous handover between satellites and ground stations. The loss of a single ground station affects Topologies B and C (inter-satellite handover to different ground stations) and may cascade to Topology D/E swarm architectures if the station serves as a Master-Satellite control point. Assessors should map each station's role in the handover topology to understand the operational impact of physical disruption.

7.3 The 'Anonymous Attacking' Problem in Physical Space

The source material identifies 'anonymous attacking' as a vulnerability in the digital domain where actors exploit the absence of national boundaries in space. An analogous physical vulnerability exists: many LEO ground stations are located in rural or semi-rural areas with minimal surveillance coverage and limited law enforcement presence. This creates conditions where physical attacks can be executed with low attribution risk. The Site Isolation / Remoteness dimension directly addresses this factor.

8. Deliverables and Reporting

Each site assessment conducted under this methodology will produce the following deliverables:

- Site Asset Register: Catalogued inventory of all physical assets with GPS coordinates and photographs.
- Context Profile Scorecard: Completed scoring across all seven dimensions with supporting evidence and source citations.
- Threat Assessment Matrix: Each threat from the taxonomy assessed for likelihood and consequence at the specific site.
- Vulnerability Gap Analysis: Current security controls mapped against identified threats, with gaps highlighted.

- Physical Security Risk Rating: Composite rating with supporting narrative and comparison to portfolio benchmark.
- Prioritised Recommendations Register: Costed and time-bound remediation recommendations ranked by risk reduction impact.

8.1 Reporting Classification

All assessment reports are classified using the Traffic Light Protocol (TLP) as mandated by the regulatory framework. Default classification for site-specific assessments is TLP:AMBER (restricted to the commissioning organisation and its direct security partners). Portfolio-level summaries may be classified TLP:GREEN for broader stakeholder distribution.

9. Quality Assurance and Methodology Governance

This methodology will be reviewed annually or following any significant change in the LEO ground segment threat landscape. All assessors must be Concentric-certified and demonstrate competence in physical security assessment, satellite communications fundamentals, and the specific regulatory environment of the country in which the assessment is being conducted.

Assessment calibration will be maintained through annual inter-rater reliability exercises where multiple assessors independently score the same site and results are compared. Maximum permissible variance between assessors on any single dimension is 1.0 points.

Appendix A: Concentric Physical Security Control Checklist

The following checklist provides the minimum control standards against which each LEO ground station will be assessed during the Vulnerability Assessment phase.

Control Area	Minimum Standard	Evidence Required
Perimeter	2.4m anti-climb fencing with topping; clear zone of 3m minimum inside and outside fence line; anti-vehicle barriers at all approach points.	Site survey / photos
Detection	Perimeter intrusion detection system (PIDS) with alarm annunciation to manned control room; response time under 5 minutes.	System specs / test logs
CCTV	Full perimeter coverage with IR/low-light capability; 30-day minimum retention; remote monitoring capability.	Camera plan / retention policy
Access Control	Multi-factor authentication at all entry points; visitor management system; vehicle search capability.	Access logs / system audit
Guard Force	Risk-appropriate manning levels; documented response procedures; training records; communications equipment.	Manning plan / training records
Lighting	Full perimeter and approach lighting to LUX standards appropriate for CCTV performance; emergency backup lighting.	Lighting survey
Resilience	UPS and backup generator with minimum 72-hour fuel autonomy; redundant fibre paths; fire suppression in data halls.	Maintenance logs / test records
Counter-UAS	For High/Critical rated sites: drone detection system with alerting; response protocol documented.	System specs / response SOP

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